

Intrusion Detection and Defense Mechanisms in Industrial Control Systems based on Deep Learning

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Abstract—With the deep coupling of the network environment and physical processes, the traditional protection methods are hard to accurately respond to the evolving and diverse intrusion threats in industrial control system (ICS). In this article the high dimensional ICS traffic data characteristic dimensionality reduction is carried out by the autoencoder for reducing the downstream network computational pressure and preserving the essential information. Instead, long short term memory (LSTM) networks were merged with the deep modeling of the dependent time series, which improved the model's capacity to identify abnormal patterns. To further enhance the recognition performance of complex attacks, a light-weighted attention mechanism is introduced to give emphasis to high-risk features on this basis. We combine the adaptive optimization algorithm with a dynamic learning rate scheduling strategy to accelerate model convergence and improve generalization and robustness. The model shows noteworthy results in term of enhancing the accuracy and reducing FPR on the Secure Water Treatment (SWaT) industrial control system dataset.

Keywords—Industrial Control System (ICS), Intrusion Detection, Autoencoder, Attention Mechanism.

I. INTRODUCTION

Industrial control systems (ICS) are the backbone of modern society and are extensively applied in major infrastructure areas, including energy, power, oil and gas, transport, and chemical industries. They are used to automate and control key industrial processes to ensure productivity, reliability and safety [1]. But ICS's critical nature also makes it a target of cyberattacks. Once attacked, it can cause production interruption and lead to economic loss, environmental pollution and personal safety. The Stuxnet worm attack on Iran's nuclear facilities, for instance, illustrates that henceforth cyberattacks can affect industrial control processes directly and bring about tangible damage.

With the continuous advance of information technology, the traditional industrial control system begins to evolve from the isolated and closed physical environment to an open and interconnect architecture. As ICS are increasingly incorporating enterprise networking, cloud computing and IoT technologies to enhance operational efficiency and enable remote

management. But the trend toward connectivity also raises cybersecurity risks significantly [2]. Attackers can gain remote access to industrial consoles by exploiting known vulnerabilities in IT systems, or launch attacks by exploiting vulnerabilities of industrial equipment, TCP/IP protocol, and industrial software security flaws.

Currently, the security protection methods of industrial control system mainly based on these popular traditional IT security technologies, such as host protection, firewalls, intrusion detection system (IDS), honeypot technology, such as to improve. Nevertheless, many limitations still exist in ICS scenarios for traditional intrusion detection approaches. For example, ICS communication protocols are very diversified, and there are big differences comparing with IT network protocols, limiting the detection capabilities of traditional IDS based on traffic analysis in industrial environments [3]. Furthermore, many industrial control systems do not tolerate delay, so traditional IT security mechanisms are often very difficult to meet ICS low latency and high reliability requirements. As well as threats from external attackers, there are, of course, the security risks of ICS networks that come from the inside, such as malicious actions by their insiders, or security risks arising from improper management.

For the past few years, deep learning-based intrusion detection technology has attracted wide attention in the research field of industrial control system security. Deep learning technology, can automatically learn and extract complex features from network traffic and system logs, adapt to the changing attack patterns, which effectively improves the accuracy and real-time performance of the detection [4]. Therefore, deep learning can not only reveal unknown attack behaviors but also lessen the dependence on human feature engineering, compared with traditional feature-matching-based detection methods.

The use of deep learning in ICS intrusion detection, however, still encounters various challenges. One such issue is the absence of label data that leads to trouble in model training, because you are trained from October 23, 2023. Furthermore, deep learning models are not interpretable, which makes the ICS

operators harder to comprehend and consequently trust. On the other hand, the deep learning model is vulnerable to adversarial attacks, which may mislead the well-designed malicious sample, and the runtime environment of the ICS has high requirements for the robustness of the model.

II. RELATED WORK

Sun et al. [5] Presented a rule-based intrusion detection system as LU-IDS (Logical Understanding Intrusion Detection System) for a better understanding of industrial control logic. LU-IDS employs tailored deep learning models to automatically intercept signatures and perform attack categorization. It also analyzes what it learns from attack classification to identify anomalous industrial control logic and generate rules. Nankya et al. [6] Describe the security for an ICS by components, protocols, industrial applications, and performance. It also shows the common threats and vulnerabilities that these systems are exposed to.

Varghese et al. [7] Industrial Control Systems (ICS) are vulnerable to malicious intrusions and thus the authors proposed a digital twin model that utilizes the digital twin of the system to improve the ICS's intrusion detection abilities. A digital twin is a virtual representation of a physical object that can model how a real world system (including a potential attack or defense) will behave.

Additionally, Wan et al. [8] described the literature on network security for the industrial control systems and described the important defense mechanisms used in these kinds of systems including intrusion detection systems (IDS), firewalls, data encryption, etc. It also examined the effectiveness of various intrusion detection methods against common types of attacks in industrial control systems such as denial-of-service (DoS) attacks, unauthorized access, and data tampering.

Zhong et al. [9] proposed a defense mechanism based on parallel RL, that seeks to improve the defense capability of ICS against multi-level attacks. Specifically, here, the game process between the attacker and defender is simulated using a reinforcement learning algorithm, and the learning process of defense strategies is accelerated through parallelization. This approach surpasses conventional single-layer defensive mechanisms as it can synchronize actions across layers of defense in response to cyber threats generated from multiple sources in real time while concurrent systems become available.

Anthi et al. [10] proposed a new three-layer IDS to enhance the security of ICS. The IDS employs a staged and supervised learning scheme for detection and response to a potential cyberattack. The first layer uses statistical traffic analysis to identify common attacks, while the second layer derives interaction data over a period of time over various devices that may result in potentially anomalous behaviors. The third layer is more about granular protocol analysis and APT/artifact detection. Zhang et al. [11] considered the robustness of deep learning-based intrusion detection systems against adversarial attacks, and proposed three protection methods. Small perturbations to input data which mislead DL models are called adversarial attacks and allow evasion of classical IDS.

III. METHODOLOGIES

A. Autoencoder dimensionality reduction and time series modeling

Initially, for the high-dimensional industrial control system (ICS) meteor data $X \in \mathbb{R}^{N \times T \times F}$, we use the autoencoder to reduce the dimensionality and feature extraction to reduce the computational burden of the subsequent network while retaining the main key information. For the original input $x_{i,t} \in \mathbb{R}^F$, the autoencoder of the autoencoder can be expressed as Equation 1, for a single sample $i \in \{1, 2, \dots, N\}$ at time step $t \in \{1, 2, \dots, T\}$:

$$z_{i,t} = \sigma(W_e x_{i,t} + b_e), \quad z_{i,t} \in \mathbb{R}^H, \quad (1)$$

where $W_e \in \mathbb{R}^{H \times F}$ and $b_e \in \mathbb{R}^H$ are trainable parameters, and $\sigma(\cdot)$ is a nonlinear activation function (such as ReLU or Sigmoid), where H represents the latent feature dimension obtained after dimensionality reduction. Subsequently, the decoder is used to map z_t back to the reconstructed output, as shown in Equation 2:

$$\hat{x}_{i,t} = \phi(W_d z_{i,t} + b_d). \quad (2)$$

This autoencoder is optimized by minimizing reconstruction errors such as mean square error (CAE), as shown Equation 3:

$$\mathcal{L}_{AE} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \|x_{i,t} - \hat{x}_{i,t}\|^2. \quad (3)$$

After the autoencoder training or pre-training, the resulting dimensionality reduction representation $z_{i,t} \in \mathbb{R}^H$ will be used as input for subsequent time series modeling, so as to retain key information and eliminate some sounds in the high-dimensional flow data.

Then, we use the Long Short-Term Memory Network (LSTM) in the time dimension to model sequence dependence and enhance the ability to capture abnormal patterns. Let $z_{i,1}, \dots, z_{i,T}$ represent the dimensionality reduction feature of a sequence, then the core state update of the LSTM at each moment t can be summarized as Equation 4:

$$h_t = LSTM(z_t, h_{t-1}), \quad (4)$$

where $h_t \in \mathbb{R}^d$ represents the hidden state of the t -th time step (omitting the detailed expansion of the gating mechanism), and d is the hidden dimension of the LSTM. By processing $z_{i,t}$ recursively, the model can retain the contextual information related to the anomaly over a long time span, laying the foundation for subsequent attack identification.

B. Attention mechanism and adaptive optimization

After obtaining the hidden state sequence $\{h_1, h_2, \dots, h_T\}$ that reflects the temporal relationship, this study further introduces a lightweight attention mechanism to assign a larger weight to potential high-risk moments or features. For an example of temporal attention, there is Equation 5:

$$e_t = v^T \tanh(W_a h_t + b_a), \quad (5)$$

where W_a , b_a and v are the trainable parameters that are used to calculate the attention energy e_t at the t -th moment. All

moments are then normalized using softmax, as in Equations 6 and 7:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{\tau=1}^T \exp(e_\tau)}, \quad (6)$$

$$\sum_{t=1}^T \alpha_t = 1. \quad (7)$$

Finally, the hidden state sequence is weighted and merged as follows in Equation 8:

$$r = \sum_{t=1}^T \alpha_t h_t, \quad (8)$$

where r is the temporal context vector after the high-risk part is highlighted. The lightweight setup allows you to capture critical information while avoiding large-scale additional calculations.

In order to output the attack detection results, let the classification layer parameters W_p and b_p , map r to the space of category C and apply softmax, as in Equation 9:

$$\hat{y} = \text{softmax}(W_p r + b_p). \quad (9)$$

In the supervised scenario, the corresponding label y can be cross-entropy loss, as shown in Equation 10:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \ln \hat{y}_{i,c}, \quad (10)$$

Combine with the autoencoder reconstruction loss $\mathcal{L}_{AE}$ or just use the $\mathcal{L}_{CE}$ alone to train the backbone network.

In the specific training process, this study uses adaptive optimization algorithms (such as AdamW or RAdam), supplemented by dynamic learning rate scheduling strategies to accelerate model convergence and enhance its generalization. In the case of AdamW, the parameter update can be summarized as Equation 11:

$$\theta_{t+1} = \theta_t - \eta_t \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda \theta_t \right), \quad (11)$$

where $\hat{m}_t$, $\hat{v}_t$ is the deviation correction value of the dynamic and variance, η_t is the current learning rate, λ is the explicit weight decay coefficient (separated from the $L2$ regular), ϵ is the smoothing term, and there is Equation 12:

$$\eta_t = \eta_0 \times \gamma^{\lfloor \frac{t}{\Delta} \rfloor} \text{ or } \eta_t = \frac{1}{2} \eta_0 \left(1 + \cos \left(\frac{\pi t}{T_{max}} \right) \right). \quad (12)$$

And other ways to achieve exponential decay or cosine annealing. By rapidly decreasing the pace in the early stage of training and smoothing the fine-tuning rhythm in the later stage, the model can converge faster and have stronger robustness in the face of changeable and high-dimensional ICS scenarios.

The proposed method can effectively identify complex attack scenarios under the combination of autoencoder

dimensionality reduction, LSTM sequence dependence, attention focusing and adaptive optimization, and significantly improve the detection accuracy and overall defense performance while maintaining a low false positive rate.

IV. EXPERIMENTS

A. Experimental setup

The experiment used the SWaT. Secure Water Treatment (SWaT) Project's A1_Dec 2015 data set, based on real-world water treatment industrial control system 7 days of normal operations and 4 days of attack are used in the dataset. This information is derived from 51 Sensors/Actuators, with multiple parameters (flow, pressure, temperature, etc.), and the data labeled accurately in normal and abnormal manner.

The SWaT dataset was chosen because it contains multiple attack scenarios against industrial control system (ICS) components, including internal and external threats, making it an ideal environment for testing intrusion detection systems (IDS). The dataset is derived from a real-world operating ICS system and contains data labeled as normal and abnormal to ensure that the model can learn from the actual operating mode.

We selected four common IDS methods for comparison:

(1) rule-based intrusion detection system, or SBIDS, which is used to detect known threats using the attack signatures;

(2) Support Vector Machine (SVM), which is a supervised learning model that separates classes by utilizing hyperplanes in high-dimensional space to categorize normal and attack behaviors;

(3) Decision Tree (DT), which selects features based on classification considerations to recursively divide data, resulting in a tree-like model;

(4) Convolutional neural networks (CNNs), a deep learning model that can automatically extract the data features and are thus fit for the complex attack patterns analysis.

B. Experimental analysis

We used five different intrusion detection methods to analyze the performance of src_mean (source IP average) under different experimental conditions. The results were presented in Figure 1 comparing the performances of the different methods given 100 experimental samples. Sensor data includes flow, pressure, and temperature measurements, which are important in ICS intrusion detection. They monitor critical operating parameters, and any deviation from normal values may indicate an intrusion or failure is occurring.

From the results of Figure 1, it can be seen that, on the whole, the Ours method is better than other methods, with small fluctuation and stability and efficiency in processing average source IP. The behaviors of other methods such as SBIDS, SVM, DT and CNN are more or less similar, with minor fluctuations, which signals that their performances are relatively inconsistent under various experimental conditions. With this comparison, a demonstration of the benefits of the Ours method in source IP data processing is achieved, particularly stability and accuracy.

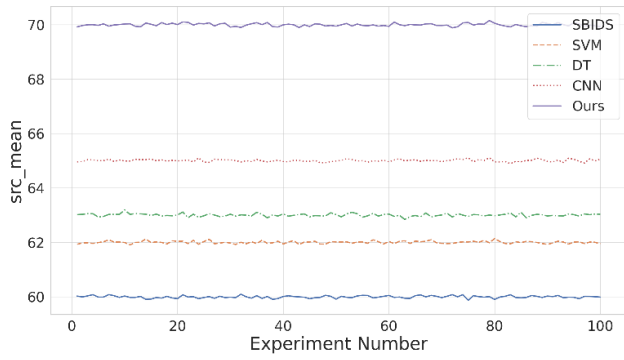


Fig. 1. Comparison of Methods on *src_mean*.

Furthermore, Figure 2 identifies the source and destination nodes of the network traffic. A strong understanding of the relationship between these two IP addresses can be a vital aspect of our network monitoring, intrusion detection, or other network security analysis.

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- 192.168.1.202 - 192.168.1.88
- 192.168.1.19 - 192.168.1.19
- 192.168.1.40 - 192.168.1.30
- 192.168.1.88 - 224.0.0.252
- 192.168.1.200 - 192.168.1.255
- 192.168.1.136 - 192.168.1.201
- 192.168.1.134 - 192.168.1.49
- 10.15.51.65 - 192.168.1.10
- 192.168.1.30 - 192.168.1.20
- 192.168.1.20 - 192.168.1.50
- 192.168.1.10 - 192.168.1.60
```

Fig. 2. Illustration of *src_ip* Unique Values Results.

The Ours method demonstrates greater stability and efficiency than other methods when processing source IP data (*src_mean*). The adaptive optimization and dynamic learning rate strategy enable the method to effectively cope with the fluctuation of source IP data while maintaining accuracy and robustness. Ours method is more consistent under different experimental conditions, and its stability and efficiency are better than traditional methods.

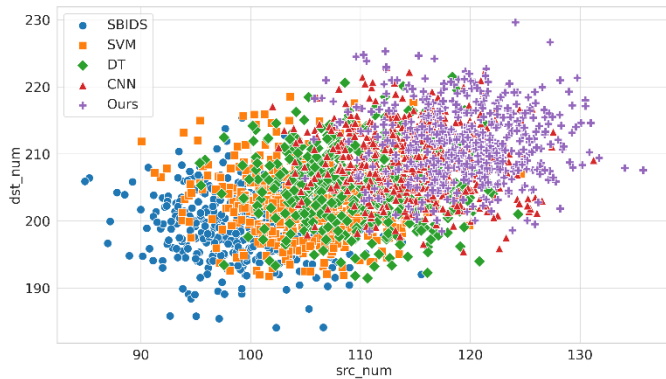


Fig. 3. Illustration of *src_num* vs *dst_num* with Method Differentiation.

Then we analyzed the distribution of 5 different intrusion detection methods for the *src_num* and *dst_num* fields to compare their performance. We employed 500 experimental

data, and plotted the distribution and discrimination of different methods on these fields.

We analyzed how the methods differ in the distribution of the data processed by the methods in Figure 3. The correlation and performance of each method between these two fields is displayed in Figure 3. In this experiment, Ours method shows clear advantages, with relatively centralized distribution and high differentiation degree, so it has good performance and stability on this task.

V. CONCLUSION

In conclusion, we studied the dependency between source IP and the target IP and assess the address distribution between various intrusion detection techniques. By visualizing these relationships, we noticed that different approaches performed differently when pairing up source and destination addresses. Our method is more consistent and capable than the other method while Bailey et al, SBIDS, SVM, DT and CNN have also demonstrated predictive power. The communication pattern between the source IP and the target IP is a significant feature to determine abnormal behaviors and security threats, these results suggest. More advanced research can flatten more network features (e.g., protocol, traffic volume, timing patterns, etc.) to process more complex pattern embeddings in network traffic.

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